

→ WINDOW FUNCTIONS

1. SECOND HIGHEST SALARY

Return the second highest salary from the table.

Hint

- This is a special case of Nth highest salary
- Use DENSE_RANK() = 2

☒ Solution

```
SELECT salary
FROM (
  SELECT
    salary,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
  FROM Employee
) t
WHERE rnk = 2;
```

#2. Find the Nth Highest Salary

Find the Nth highest salary from an Employee table.

Table Structure

Column Name	Type
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id	INT
----	-----

name	VARCHAR
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salary	INT
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☒ Solution

```
SELECT salary
FROM (
  SELECT
    salary,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
  FROM Employee
) t
WHERE rnk = n ;
```

#3. Running Total (Cumulative Sum)

Calculate a running total of sales by date.

Table Structure

Column Name Type

sale_date DATE

amount INT

Solution

```
SELECT
  sale_date,
  amount,
  SUM(amount) OVER (ORDER BY sale_date) AS running_total
FROM Sales;
```

4. Rolling (Moving) Average – 7 Days

Calculate a 7-day rolling average of daily sales.

Table Structure

Column Name Type

sale_date DATE

amount INT

Solution

```
WITH cte AS (
  SELECT
    sale_date,
    amount,
    ROW_NUMBER() OVER (ORDER BY sale_date) AS rn
  FROM Sales
)
SELECT
  sale_date,
  amount,
  AVG(amount) OVER (
    ORDER BY sale_date
    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW
  ) AS moving_avg_7_days
FROM cte
```

```
WHERE rn >= 7;
```